Don’t be THAT GUY using AI

Professional work when AI writes the first draft

Michael Tomlinson · Tyto Insights

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The spec he sent the client

Generated with AI Assistant

CampusPass: A Comprehensive Solution for Seamless Event Ticketing — Empowering Student Societies in the Digital Age

A Holistic Technical and Strategic Overview for Stakeholders

1. Executive Summary

Setting the Stage for Transformational Success

In today's rapidly evolving digital landscape, student societies face a myriad of challenges when it comes to organizing, promoting, and managing events — challenges that range from ticket distribution and payment processing to attendee verification and post-event analytics. CampusPass is a cutting-edge, end-to-end ticketing platform designed to address these pain points head-on, delivering a seamless, intuitive, and robust experience for all stakeholders involved. By leveraging modern cloud-native technologies, a microservices-based architecture, and industry-leading best practices, CampusPass empowers societies to effortlessly create events, configure flexible ticket tiers, and sell tickets — while simultaneously enabling students to discover, purchase, and manage their tickets with unprecedented ease. Furthermore, the platform's innovative QR-based check-in system ensures a frictionless entry experience, eliminating the need for paper tickets and manual verification processes. It is important to note that CampusPass is not merely a ticketing tool; rather, it is a holistic ecosystem that fosters community engagement, drives event attendance, and unlocks valuable insights through comprehensive analytics dashboards. Moreover, the platform has been designed with scalability at its core, ensuring that it can seamlessly accommodate everything from intimate society gatherings to large-scale campus-wide festivals with thousands of concurrent users. In addition, robust security measures — including end-to-end encryption, role-based access control, and compliance with relevant data protection regulations — ensure that sensitive user data remains protected at all times. Ultimately, CampusPass represents a paradigm shift in how student societies approach event management, positioning them for sustained success in an increasingly competitive and digitally driven environment. This document provides a comprehensive overview of the platform's architecture, features, implementation strategy, and long-term vision, serving as a foundational reference for all stakeholders involved in the project's success.

The code he wrote

EXPLORER · CAMPUSPASS

- ▾ campuspass
  - app.py — 5,214 lines
  - app_backup.py
  - app_v2_FINAL.py
  - app_v2_FINAL_fixed.py
  - test_app.py
  - utils.py
  - helpers.py
  - helpers2.py
  - .env
  - .env.local
  - .env.prod
  - .env.prod.backup
  - requirements.txt
  - README.md

app.py | test_app.py | .env

```
3118  # Initialize the variable
3119  total = 0
3120  API_URL = "http://localhost:3000"  # change in production
3121  MAX_TICKETS = 500
3122  ADMIN_PASSWORD = "campuspass123"
3123
3124  def process_ticket_purchase(data):
3125      # Process the ticket purchase
3126      # TODO: Add auth method here
3127      global total
3128      total = total + 1  # increment total by 1
3129      if total <= MAX_TICKETS:  # check if total is less than or equal to max
3130          return charge_card(data["card"], 180)  # price
3131      # Insert error handling here
```

The email he sent

To Lerato Mokoena (Student Societies Council)
From Sam (Tyto Insights)

Important Update Regarding the Upcoming CampusPass Demonstration Session 🗓️

Dear Lerato,

I hope this email finds you well and that your week is off to a fantastic start! I wanted to take a moment to reach out regarding the upcoming CampusPass demonstration session that we had previously scheduled together.

📌 Background

As you know, our team has been working diligently on developing the CampusPass platform — a comprehensive solution designed to streamline event ticketing for student societies. We have made significant progress across a number of key areas, including ticket purchasing, QR-based check-in, and the society admin dashboard, and we are very excited to showcase these features to you and the wider council.

🔄 Update to the Schedule

In order to ensure that we are able to deliver the most polished and seamless demonstration possible, and after careful consideration of a number of factors, we have made the decision to move the demonstration from Tuesday to Thursday at the same time (14:00). We believe this adjustment will allow us to present a more robust and complete experience.

Now picture this person.

The 30-minute polish

01

10+ hrs

Spec written by hand

02

10 min

AI draft: THAT GUY sends this

03

+30 min

High-quality work

Who’s talking

Michael Tomlinson

Final-year BSc Computer Science, University of Pretoria

Tyto Insights

AI Engineer, working on systems that use AI heavily

AI glossary
tytoltd.github.io/AI_PROFESSIONALISM

01

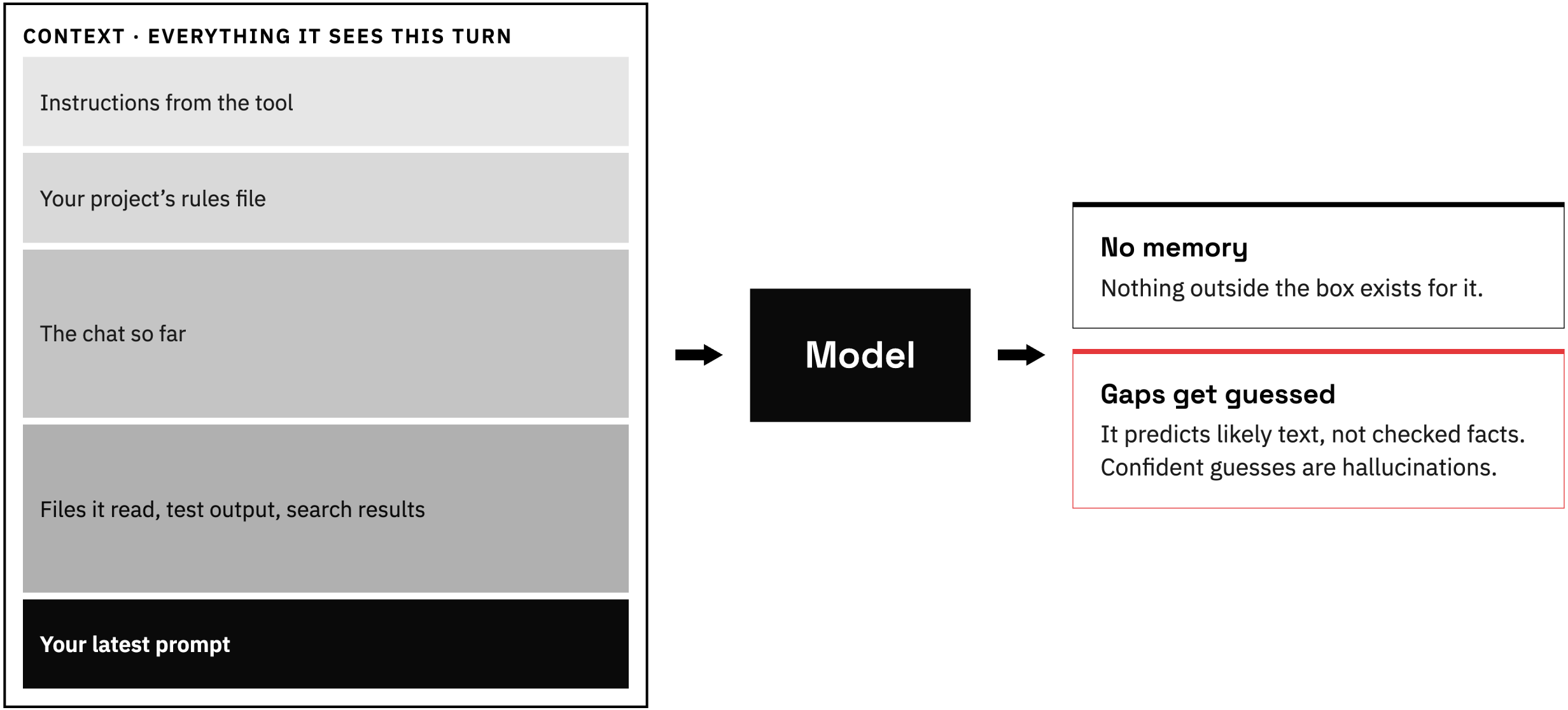
Context

What it knows, and how long it stays sharp

01	It has the depth, not your context
02	If it isn’t in the box, it guesses, confidently
03	Long sessions degrade, even with huge context windows
In action	One task per session, the plan in a file

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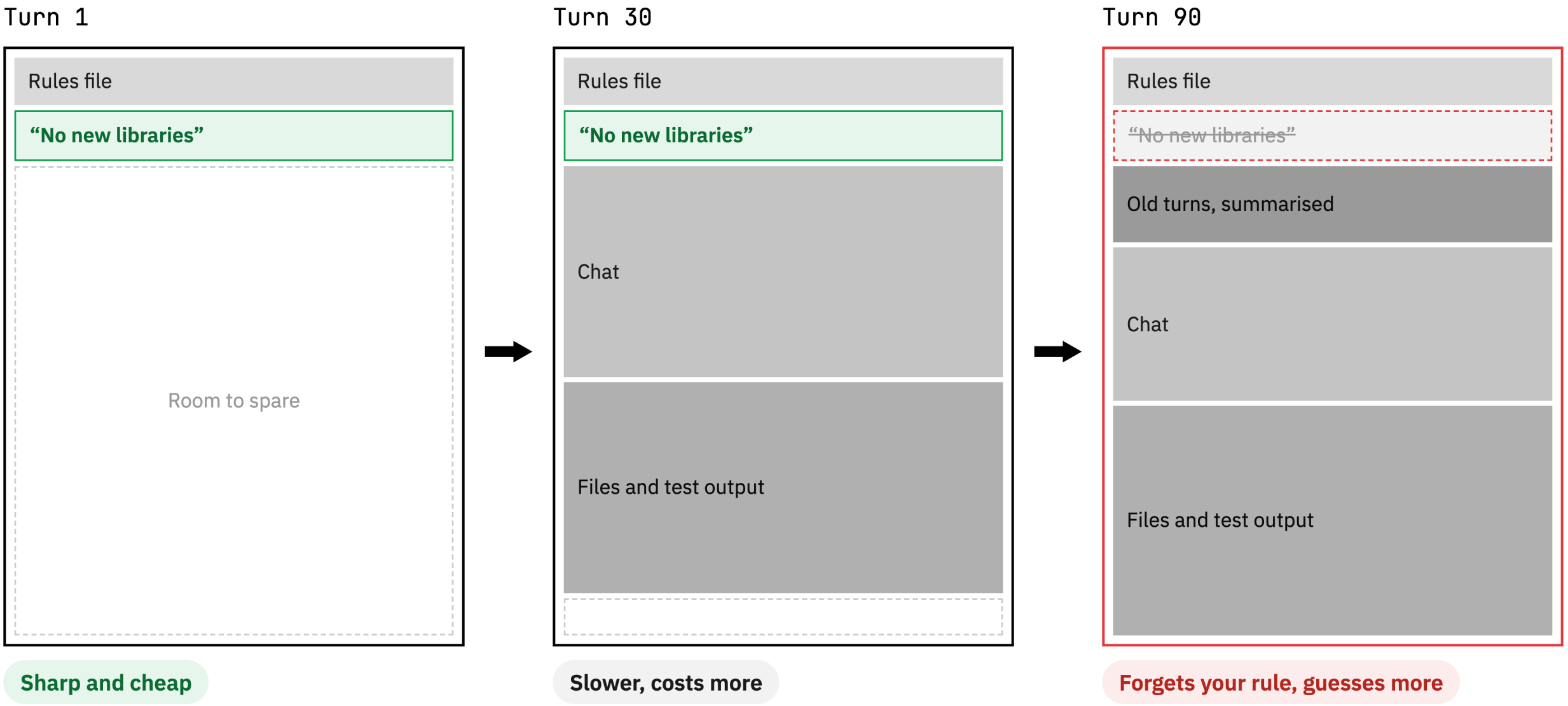
It only knows what's in the box



The harness lets it act, in a loop



Long sessions degrade



One task per session. The plan lives in a file.

Session 1 · closed
Built phase 1

Session 2 · closed
Built phase 2

Session 3 · now
Reads the plan, builds phase 3

PLAN.md

```
# Ticketing proof of concept

[x] Phase 1: buy a ticket
    gate: test_buy passes

[x] Phase 2: QR ticket after payment
    gate: test_qr passes

[ ] Phase 3: scan at the door
    gate: test_scan passes

## Notes for the next session
Payments are faked in fake_pay.py
```


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02

Planning

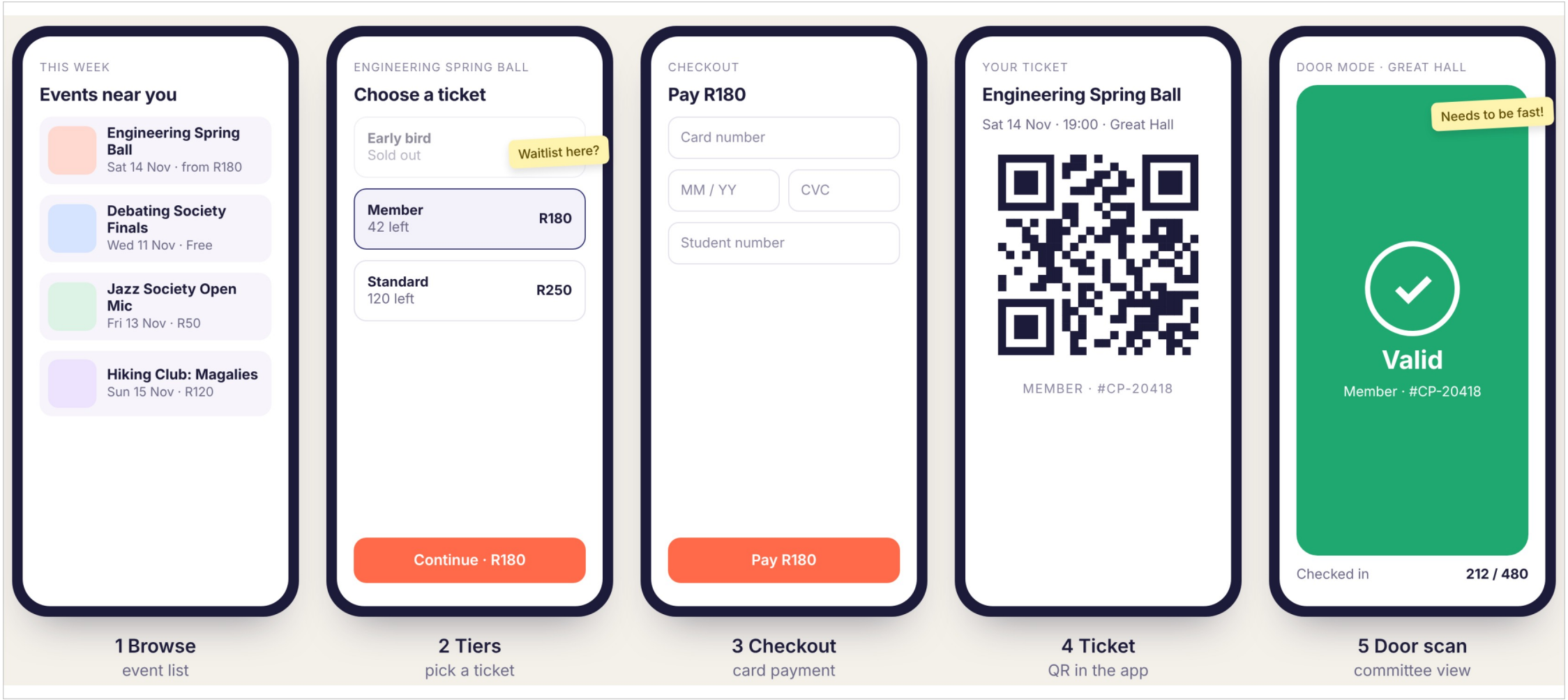
Before any real code

01	Settle style and flow with a throwaway prototype
02	Tell it the scale: prototype, proof of concept, internal tool or product
03	Colleague test: could someone new build it from your prompt?
In action	Shared vocabulary: say “Strategy”, ask for options, know the cons

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A throwaway prototype settles style and flow

FIVE CLICKABLE SCREENS · FAKE DATA · ONE AFTERNOON



A vague prompt gets guesses

Build a ticketing app for student societies.

Done! Here's your full ticketing platform:

- Login, roles and an admin panel **guess**
- Live card payments **guess**
- Email, SMS and push alerts **guess**
- An analytics dashboard **guess**

`app.py · 5,214 lines` **THAT GUY's file**

A clear brief gets questions

Scale: proof of concept for a demo next week, then thrown away.
Goal: issue a QR ticket after payment, and scan it.
Stack: Python, FastAPI, SQLite. No login, because it's thrown away.
Done when: a test that buys and scans one ticket passes.
Ask me questions first.

A few questions before I start:

1. Should I fake the payment step?
2. Can a ticket be scanned twice?
3. Is a terminal demo enough, or do you need a page?

It asks instead of guessing

Vague words get vague code

Make pricing flexible.

Sure, I've updated price():

```
def price(ticket, user, group):
    if ticket.kind == "early":
        if user.is_member:
            return 80
        elif group > 5:
            return 70
    elif ticket.kind == "member":
    # ...11 more branches
```

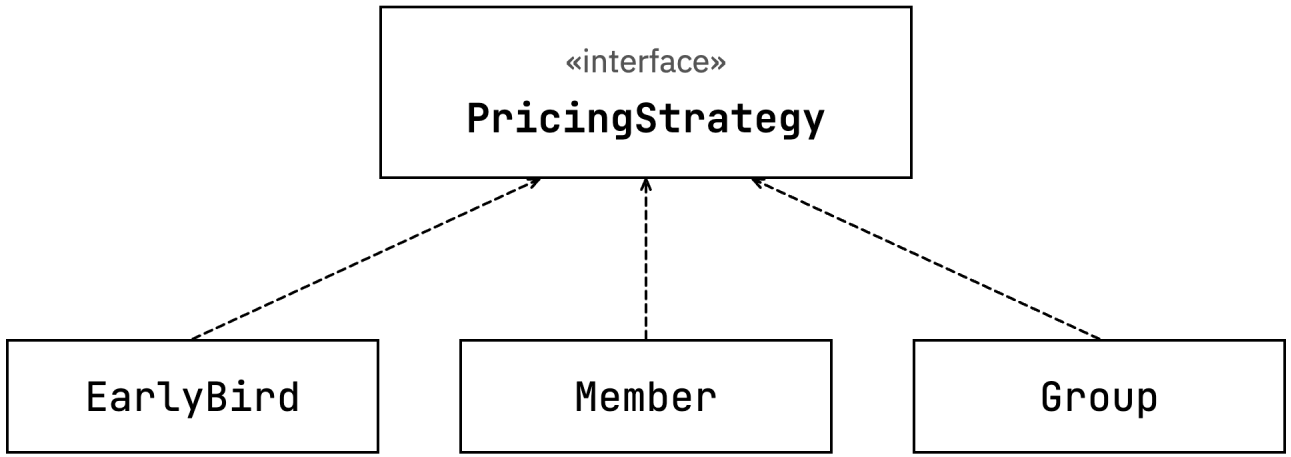
Every new ticket type edits this function

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Shared vocabulary: speak design patterns

Use the Strategy pattern for pricing.

Here’s the design:



Con: more classes to keep track of. Worth it if ticket types keep changing.

A new ticket type is one new class

03

Code

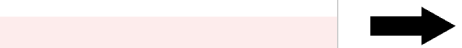
What it learns from your codebase

01	It copies what it sees, good or bad
02	Show it good code. Say what to do, not only what not to do
03	One command to build, test and lint. Warnings fail
In action	A rules file, read every session. Add a line when it repeats a mistake

It copies what it sees, good or bad

refunds.py — yours, written in a rush

```
def refund(order):
    try:
        pay.refund(order.id)
    except:
        pass
    print("refunded", order)
```



waitlist.py — new, written by AI

```
def notify(user):
    try:
        email.send(user.addr)
    except:
        pass
    print("notified", user)
```

One shortcut

Now it’s the house style

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Write it down once: the rules file

CLAUDE.md

```
# Commands
make check  # build, test, lint. Warnings fail

# Code
No hard-coded passwords or URLs. Use config.
Many cases? Use a switch, not an if/else chain.

# Tests
Every test checks real behaviour. No assert True.

# Don't touch
payments/ without asking first
```

New line: it wrote assert True twice, so now it's a rule



In the box for every session



One command to check everything



Add a line when it repeats a mistake

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04

Testing

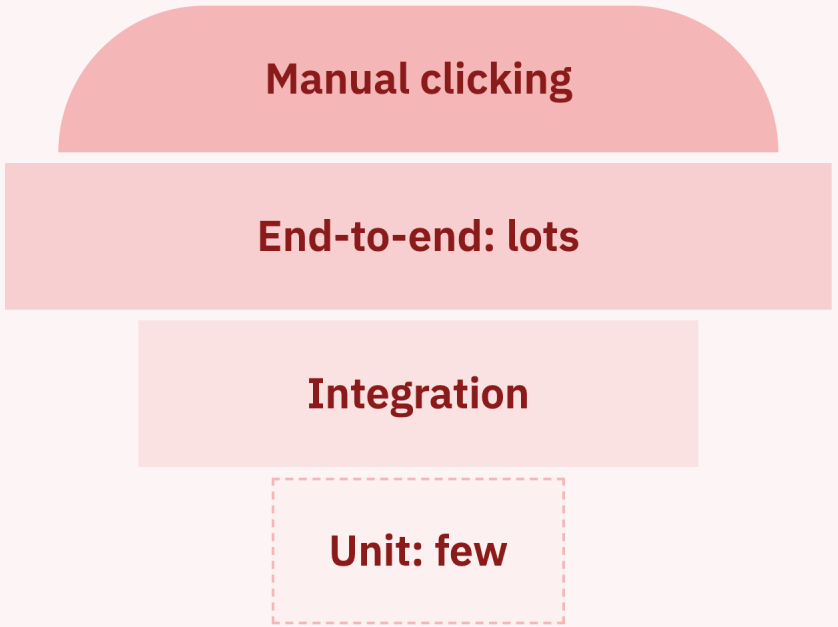
How you know the code works

01	Tests are what let you trust the code
02	The wrong shape, an ice-cream cone, makes CI slow and flaky
03	Tests are where agents cheat the most
In action	Say exactly which tests to write, at each level. A fresh agent reviews

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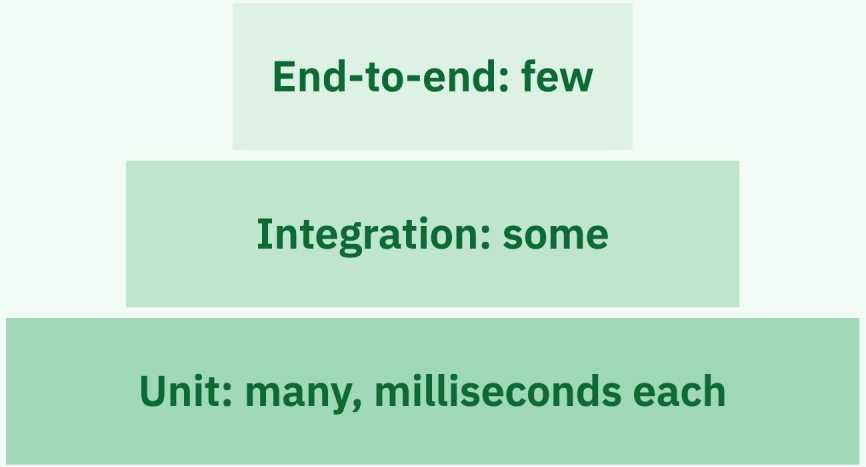
Get the shape right: a pyramid, not an ice-cream cone

UNGUIDED AI: THE ICE-CREAM CONE



Slow, flaky CI

WHAT YOU ASK FOR: THE PYRAMID



Fast CI you can trust

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“Make the tests pass”

Pull request · fix: tickets no longer oversell — ✓ All checks passed

`tests/test_checkout.py` — 01 · Deleted the failing test

```
- def test_no_oversell():
-     buy(buyers=50, tickets=10)
-     assert sold() == 10
```

`tests/test_load.py` — 02 · Loosened the check

```
-     assert sold() == 10
+     assert sold() ⩽ 12
```

`checkout.py` — 03 · Faked the code it tests

```
  def reserve(ticket):
-     with db.lock(): ticket.left -= 1
+     return True
```


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Say exactly which tests to write

Write tests for checkout().

Unit — **Edge cases:** 0 tickets left, the last ticket, 2 buyers at once, a refund after sell-out

Integration — **Error handling:** card declined, payment times out, database down

End-to-end — **One happy path:** buy, get the QR, scan it at the door

Don’t change existing tests. Don’t fake reserve().

Plan: 9 unit, 3 integration, 1 end-to-end. Writing them now.

A pyramid, not a cone

Agents judge each other's work better than their own

THE SESSION THAT WROTE IT

Review your changes.

Looks good!

- ✓ Code is clean
- ✓ All 214 tests pass
- ✓ Ready to merge

Marks its own work

A FRESH SESSION, GIVEN THE SPEC

Review this branch against SPEC.md.

Found 3 problems:

`assert True`

This test checks nothing. It always passes.

`DB_PASSWORD = "admin123"`

Password hard-coded. Move it to config.

`if … elif … elif …  (40 branches)`

A switch would be far easier to read.

Catches what the author missed

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05

Communication

How the work leaves your hands

01	Clear writing respects the reader’s time
02	Low-density writing gets skimmed. AI tells signal low effort
03	Run it before you hand it over
In action	Your bullets, AI’s paragraph, your cut. One fact first

Your bullets, AI's paragraph, your cut

1 · YOUR BULLETS

- Scanning works offline
- Duplicate scan: reject, show first scan
- Door role only, own society
- 500 people in 20 min, 4 phones

2 · AI EXPANDS IT · 120 WORDS

In today's fast-paced event landscape, ensuring a seamless and robust check-in experience is paramount. CampusPass empowers door staff to validate tickets offline, ensuring continuity of operations even in challenging connectivity environments. Duplicate scans are rejected, with details of the original scan displayed to foster trust and integrity. Access is restricted to Door-role accounts for each society, leveraging industry-leading role-based access control. With a target of 500 attendees in 20 minutes across 4 devices, CampusPass delivers a frictionless entry experience.

3 · YOU CUT IT · 38 WORDS

Door staff scan tickets on their own phones, even without signal. A duplicate is rejected and shows who scanned it first.

Only Door-role accounts can scan, for their own society. Target: 500 people in 20 minutes on 4 phones.

Where’s the one fact?

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From Sam (Tyto Insights)

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Lead with the one fact

To Lerato Mokoena (Student Societies Council)
From Sam (Tyto Insights)

CampusPass demo moved to Thursday 14:00

Hi Lerato,

We've moved the CampusPass demo from Tuesday to **Thursday at 14:00**, same venue. The extra two days let us finish the door-scanning screen so you can try it yourself.

Does Thursday work for you and the council?

Thanks,
Sam

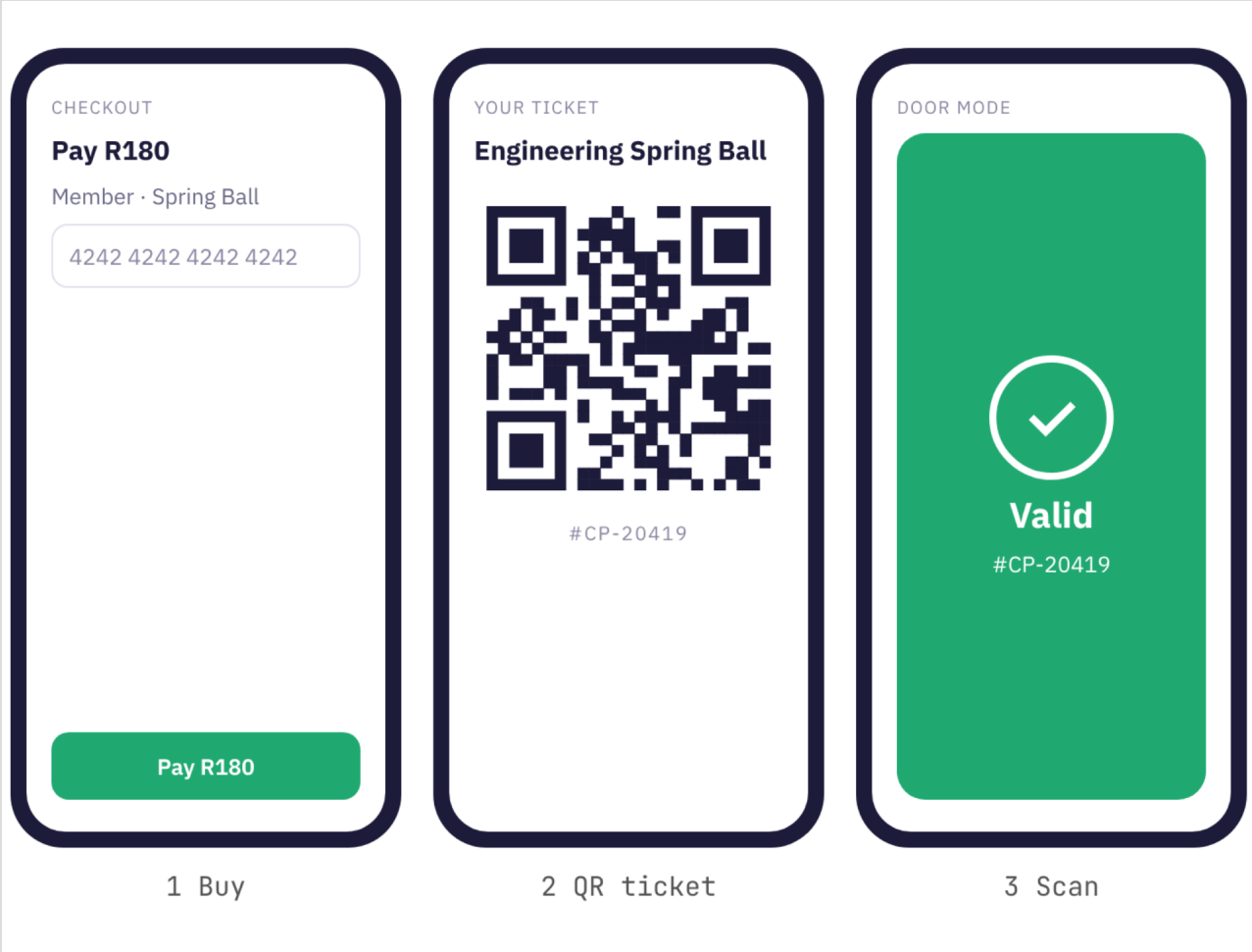
The date is in the subject line

One question, one-word answer

No emoji, no filler

Never hand over work you haven’t run

1 · RUN THE HAPPY PATH YOURSELF



2 · SAY WHAT CHANGED, IN THREE LINES

Cause

Two buyers could both get the last ticket.

Fix

The ticket count now updates in one step. The losing buyer sees “Sold out”.

How I tested it

50 buyers, 10 tickets, exactly 10 sold. Bought and scanned one myself.

AI is your intern.

Takeaways

01 Context	02 Planning	03 Code	04 Testing	05 Communication
One task per session	Prototype, then throw it away	Show it good code to copy	Ask for a pyramid: which tests, at which level	Your facts, your cut
Keep the plan in a file	Say the scale	One command to check it all	Read the tests it changed	One fact first
Put your intent in the box	Ask for options, in pattern terms	Keep a rules file	A fresh agent reviews	Run it before you hand it over

Glossary and notes · tytoltd.github.io/AI_PROFESSIONALISM

Questions?

Glossary and notes
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